Evaluation Evidence

Measured accuracy, FPS, latency, liveness results and documented failure cases for the VisionGuard multi-module CCTV analytics pipeline. Test dates: 26–27 October 2026.

How to read this document. Every number below comes from a real model run on real video or real enrollment photos, saved as JSON under `runs/`. Where no independent ground truth exists, the metric is reported as **NOT MEASURED** — never as zero or perfect. Small samples are labelled as such; they show the pipeline works end-to-end, not field-certified accuracy.

1. Results at a glance

Area	Headline result	Evidence size	Status
Face recognition (enrolled users)	**98.7%** precision, **92.8%** recall; 1 wrong name in 414	420 photos, 14 identities, 193 templates	**Measured**
Person detection + tracking	8 TP / 0 FP / 2 FN (P 1.00, R 0.80); 0 ID switches, 0 fragments	2 labelled CCTV frames, 10 people	**Small sample**
Safety events (fall, abandoned bag, gate, tailgating)	Precision 1.00 / recall 1.00, 0 false events	1 staged event each + 4-person crossing	**Small sample**
PPE (safety vest)	4 TP / 0 FP / 3 FN (P 1.00, R 0.57)	1 factory frame, 7 vests	**Weak recall**
Cross-camera Re-ID	0 / 1 positive pair matched	1 URFall actor, 2 cameras	**Failing**
Throughput	4.8 – 28.2 FPS end-to-end per scenario; 4,013 frames, **0 failed**	12 scenarios	**Measured**
Latency	Person detector 32.5 ms mean / 38.9 ms p95; full face pipeline 77.8 ms mean	per-module, warm-up excluded	**Measured**
Face liveness	5/5 behaviour tests pass; anti-spoof accuracy not measured; false spoof flags seen live	unit tests + ~90 s live CCTV	**Experimental**

Test environment

GPU	CPU / RAM	Software	Conditions
NVIDIA GeForce RTX 5060 (CUDA 13.0)	AMD Ryzen, 12 logical cores / 16 GiB	Python 3.14.7, torch 2.14.1+cu130, ultralytics 8.4.161, OpenCV 5.0.0.93, OpenVINO 2026.4.1	Live camera workers kept running during tests: shared-load numbers, not an isolated capacity test.

2. Accuracy

2.1 Face recognition against enrolled users

Read-only run of the configured YOLO-face detector + SFace embedder (threshold 0.38, margin 0.05) over every export photo of the active registry. Detection failures are counted as misses.

Metric	Value	Count
Correct identity (known people)	92.8%	384 / 414
Precision (named results that were right)	98.7%	384 / 389
Recall	92.8%	384 / 414
False reject rate (known person shown as unknown)	7.0%	29 / 414
Wrong-identity rate	0.2%	1 / 414
False accept of unregistered labels	66.7%	4 / 6
Near-identical template/photo overlap	0	checked

Scope: enrollment-export photos, not independent CCTV footage. The 6 'unknown' rows are labels absent from the registry and may be aliases or mislabels; this sample is too small to rate unknown-person rejection.

2.2 Detection, tracking and event modules

Module	Scope scored	Precision	Recall	FP	Other
Person detection	Boxes, 2 CCTV frames (IoU ≥ 0.5)	1.000	0.800	0	8 TP, 2 FN
Person tracking	Same frames, track IDs	1.000	0.800	0	0 ID switches, 0 fragments
Fall detection	Event window, URFall fall-01	1.000	1.000	0	No false 'recovered' event after fix
Abandoned object	Event window, 2,189-frame public clip	1.000	1.000	0	20 s unattended rule
Gate counter	4-person crossing group	1.000	1.000	0	4 entries, 4 line crossings
Tailgating	Same group	1.000	1.000	0	3 tailgating events
PPE (Safety Vest only)	Boxes, 1 factory frame	1.000	0.571	0	4 TP, 3 FN (distant workers)
Person Re-ID (cross-camera)	1 actor, 2 cameras	n/a	0.000	0	0/1 matched at threshold 0.6
Re-ID (same camera, descriptor)	4 queries	—	—	0	Top-1 1/4; TP 1, FN 3, TN 12
ANPR, weapon, crowd, loitering	No ground truth	—	—	—	**NOT MEASURED**; ran without errors (see 2.3)

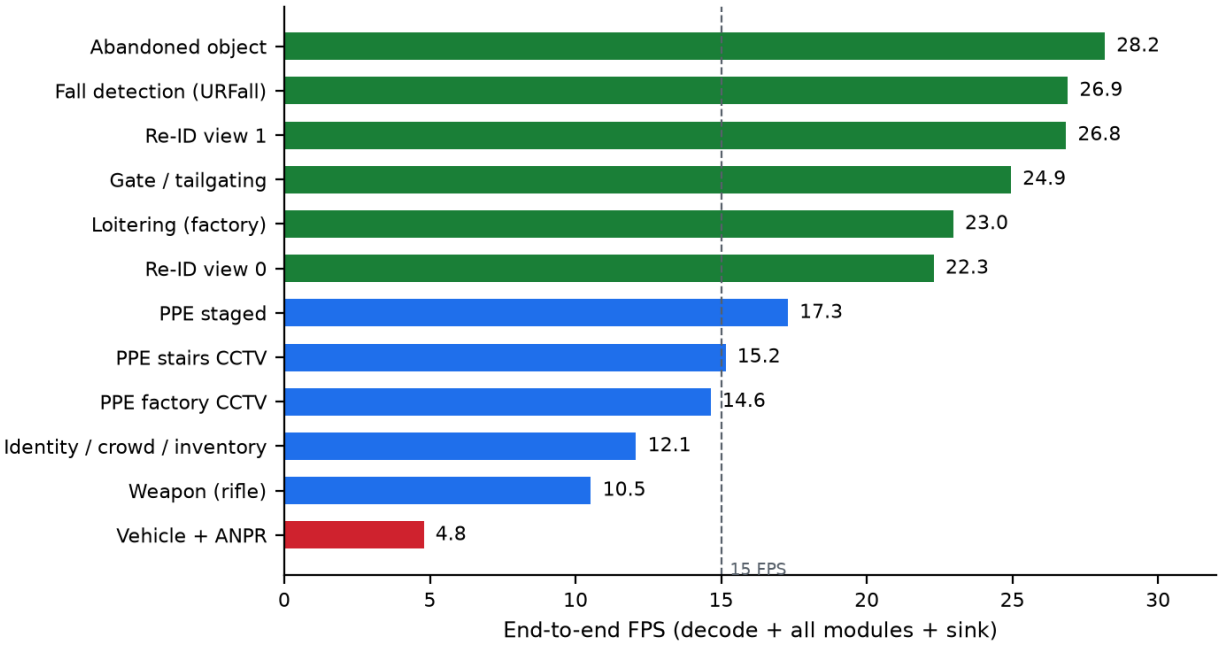
Matching: same-class Hungarian assignment at IoU ≥ 0.5; unlabelled frames are never counted as false positives. Perfect scores on one staged event show the path works, not field reliability.

2.3 Events observed on real video (no ground truth)

Scenario	Frames	Events produced
Vehicle + ANPR	80	3 plate_read (304 plate detections in 79/80 frames)
Weapon (rifle clip)	80	1 weapon_detected, 1 armed_person (weapon boxes in 10/80 frames)
Identity / crowd / inventory CCTV	120	1 unknown_person, 1 attendance_check_in, 2 inventory_low
Staged PPE video	600	2 ppe_violation (both reached dashboard alerts)
Factory loitering (5 s eval threshold)	102	5 loitering
Fall monitoring (fall-detection-data.mp4)	236	1 fall_detected; person kept marked frames 85–236

3. FPS (throughput)

End-to-end FPS includes video decode, every enabled module, sink collection and cleanup; model loading is excluded. Each scenario runs only the modules it needs, so heavier stacks are slower.



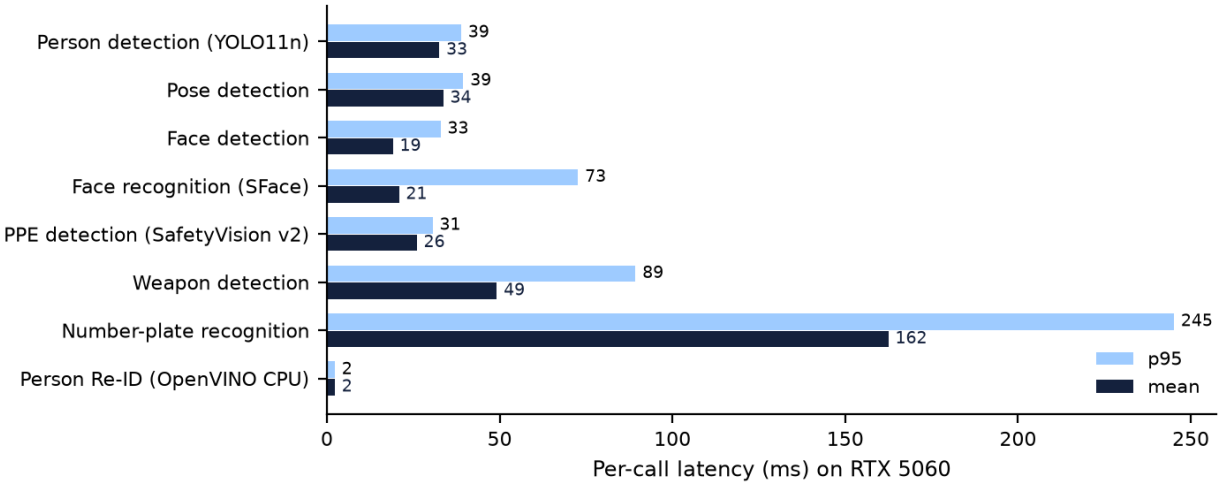
Green ≥ 20 FPS, blue 10–20 FPS, red < 10 FPS. ANPR is the bottleneck at 162 ms per call.

Scenario	Frames	Failed	End-to-end FPS
Abandoned object	2,189	0	28.17
Fall detection (URFall)	160	0	26.90
Re-ID view 1	160	0	26.84
Gate / tailgating	140	0	24.95
Loitering (factory)	102	0	22.97
Re-ID view 0	160	0	22.30
PPE staged	600	0	17.29
PPE stairs CCTV	120	0	15.15
PPE factory CCTV	102	0	14.64
Identity / crowd / inventory	120	0	12.08
Weapon (rifle)	80	0	10.53
Vehicle + ANPR	80	0	4.79
Total	**4,013**	**0**	4.79 – 28.17

Deployment check (27 Oct): CPU 40 frames, 0 failed, up to 5 tracked people; GPU 40 frames, 0 failed, up to 5 tracked people. Behind nginx, five simultaneous MJPEG streams each returned HTTP 200 with frames received.

4. Latency

Per-call latency after 10 warm-up frames. Shared detector cost is charged to its first consumer, so downstream rules look nearly free. Inverse-call FPS is not multi-camera capacity.



Model-backed modules only (rule modules are under 1 ms and omitted from the chart).

Module	Mean ms	p95 ms	Inverse-call FPS	Scenario
Person detection (YOLO11n)	32.54	38.86	31	abandoned-final
Pose detection	33.66	39.47	30	fall-final
Face detection	19.10	32.89	52	identity-cctv
Face recognition (SFace)	20.87	72.68	48	identity-cctv
PPE detection (SafetyVision v2)	26.01	30.59	38	ppe-v2-cctv
Weapon detection	49.17	89.30	20	weapon-final
Number-plate recognition	162.48	245.14	6	vehicle-anpr
Person Re-ID (OpenVINO CPU)	2.30	—	435	reid-cam0
Low-light check	0.88	1.22	1,136	identity-cctv
Person tracking	0.18	0.27	5,556	identity-cctv
Fall rule	0.11	0.17	9,091	fall-final
Tailgating / gate rules	0.06	0.07	16,667	tracking-gate
Face pipeline on photo (decode→detect→embed→match)	77.84	100.59	12.8	420 enrollment photos

Re-ID p95 is not shown because the module embeds only on some frames, so its p95 call is a cache hit (0.06 ms) and would mislead.

5. Face liveness

Passive heuristic that runs before recognition: per face track it looks for an eye blink or non-rigid (3D) landmark motion. A flat photo moves by one 2D similarity transform; a real head does not. It outputs *live*, *checking* or *spoof* and raises one `spoof_suspected` alert per spoof track.

5.1 Behaviour tests (re-run 27 Oct 2026: 5 passed)

Test	Expected	Result
A flat photo moves rigidly, a head does not	Photo residual ≈ 0; head residual high	**PASS**
Still photo after decision time (4 s)	state = spoof, one spoof_suspected event	**PASS**
Head moving in 3D	state = live, no events	**PASS**
Face seen for less than the decision time	state = checking (recognition still allowed)	**PASS**
Recognition with block_spoof on	Spoof face is not named	**PASS**

5.2 Calibration measurements on camera

Signal	Printed/phone photo (moved, tilted, zoomed)	Threshold set	Real person
Landmark 3D residual	0.04 (max observed)	> 0.06 = 3D motion	Moving head exceeds threshold
Eye-patch change (blink)	0.58 (max observed)	> 0.70 and reopen within 2 frames	Blink exceeds threshold
Landmark jitter at 640 px	median 0.027 (moved photo)	—	median 0.032 (standing still)
Moiré / texture peak	High	Off by default	Real CCTV faces scored equally high

5.3 Live CCTV observation and limits

• In about 90 seconds across four live cameras, **2 real people were flagged as spoof** (they stood still and did not blink in view). That is why `block_spoof` is **off by default**: the alert is a 'check this' flag, not a gate.
• A still photo and a still real face overlap at 640 px detection size (0.027 vs 0.032 jitter), so passive cues cannot reliably separate them.
• A replayed video or a good 3D mask will pass. No anti-spoofing dataset (e.g. attack/bona-fide APCER/BPCER) was run: **anti-spoofing accuracy is NOT MEASURED**.
• Planned fix: a trained anti-spoofing model or depth/IR camera.

6. Failure cases

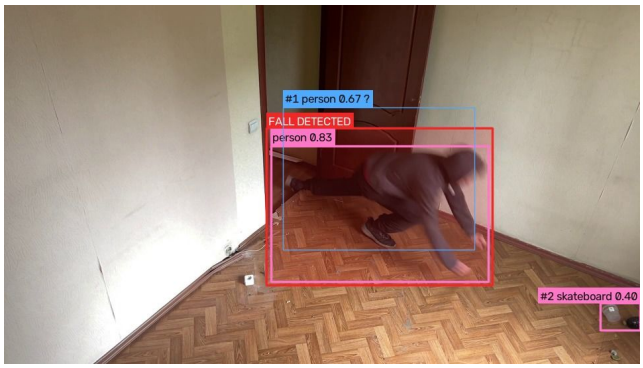
Failures found during evaluation, with what was done about each. Fixed items were reproduced by a failing regression test before the fix and pass after it.

#	Failure	Evidence	Cause	Status
1	PPE misses distant workers' vests	3 of 7 vests missed on factory frame (recall 0.57)	Small, far objects at CCTV resolution; crop-upscaling experiment did not help	Open
2	Cross-camera Re-ID missed the same person	0/1 positive pair matched; same-camera top-1 1/4	Appearance embedding differs across views; threshold kept at 0.6 rather than lowered to pass	Open
3	Face: known people rejected as unknown	29 of 414 photos (7.0%)	Pose, blur, small faces below quality gates	Open
4	Face: unregistered labels accepted under a registered name	4 of 6 rows	Possibly aliases/mislabels in export; needs label review	Under review
5	Face: one wrong name	1 of 414 photos	Look-alike template overlap	Open
6	Liveness flags still real people	2 real people flagged in ~90 s, 4 cameras	Passive cues cannot separate still face from photo	Mitigated: blocking off
7	False 'fall recovered' when pose track was lost	First real fall run emitted recovery immediately	Track loss treated as recovery	Fixed
8	Abandoned bag alert lost between frames	Detection gaps; class flips backpack/handbag/suitcase	Tracker dropped low-score continuations	Fixed
9	Re-ID gallery lost one of two identities	Two processes writing at once kept only one	Read/merge/write race	Fixed (file lock)
10	ANPR is slow	162 ms mean, 245 ms p95; scenario at 4.79 FPS	Plate detector + OCR per vehicle	Open
11	Weapon seen on only 10 of 80 frames	Rifle clip	Occlusion and pose; event still raised once	Partial
12	One regression test fails	748 passed, 1 failed	Test expects identity modules disabled; current config enables face modules (intentional)	Known

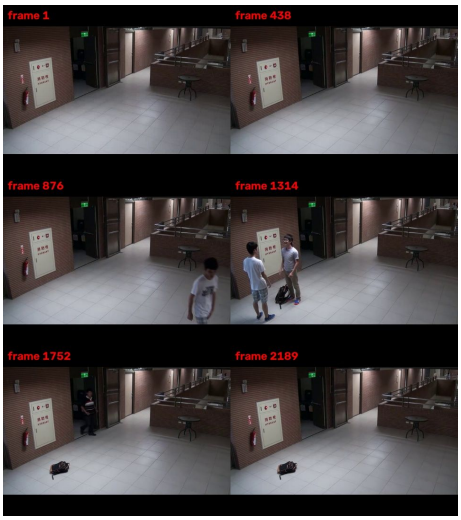
6.1 Visual evidence



PPE v2 on factory CCTV: near vests found, distant workers missed (failure 1).



Fall detected and the person kept marked until upright (after fix 7).



Abandoned-bag clip, reviewed frames 1 → 2189 (fix 8).



Rifle detected and armed_person raised (failure 11: sparse frames).

7. Limitations

• Sample sizes are small (one staged event per safety module, 2 labelled tracking frames, 1 PPE frame). Treat 1.000 scores as 'path works', not as general precision/recall.
• The PPE calibration frame was also used for model selection; training-set overlap is unknown.
• Face accuracy is on enrollment-export photos, not independent CCTV identities.
• FPS/latency were measured with production camera workers running; they are not a five-camera capacity guarantee.
• Loitering used a 5-second evaluation threshold, not the 30-second production default.
• Still needed: independent named CCTV labels, dense tracking annotation, more positive/negative safety clips, a camera-pair Re-ID set, and an anti-spoofing dataset.

8. Reproduce

```
.venv\Scripts\python.exe -B -m tools.benchmark_cv --manifest
runs/module-validation-20261026/final-scenarios.json --output runs/recheck-final
.venv\Scripts\python.exe -B -m tools.benchmark_cv --manifest
runs/module-validation-20261026/ppe-v2-scenarios.json --output runs/recheck-ppe
.venv\Scripts\python.exe -B -m tools.benchmark_cv --manifest runs/module-validation-20261026/reid-scenarios.json
--output runs/recheck-reid
.venv\Scripts\python.exe -B -m tools.evaluate_recognition --registry data/faces/faces.db --photo-manifest
"data/faces/New user face data/face-photos/manifest.csv" --config configs/config.yaml --json-output
runs/recheck-face.json
.venv\Scripts\python.exe -m pytest tools/tests/test_mvp_features.py -rA
```

Evidence files

File	Contains
runs/module-validation-20261026/BENCHMARK.md	Per-module latency / FPS / precision / recall tables
runs/module-validation-20261026/combined-benchmark.json	Machine-readable results for all 12 scenarios
runs/module-validation-20261026/face-registry-diagnostic.json	All 420 face predictions and metrics
runs/module-validation-20261026/reid-cross-camera.json	Cross-camera Re-ID failure
runs/module-validation-20261026/*-truth.json	Independent ground-truth annotations + provenance
runs/module-validation-20261026/run-metadata.json	Hardware, software versions, test conditions
runs/deployment-validation-20261027/	CPU/GPU pipeline and nginx proxy checks
computer_vision/modules/recognition/face_liveness.py	Liveness method and calibration notes